

```
enum class Visibility { VISIBLE, INVISIBLE; }
```

```
class Color {
```

```
private:
```

```
    unsigned char r, g, b;
```

```
public:
```

```
    static Color red;
```

```
    static Color green;
```

```
    static Color blue;
```

```
    Color(unsigned char r, unsigned char g, unsigned char b);
```

```
    Color(uchar cyan, uchar magenta, uchar yellow, uchar black);
```

```
    string toString() const;
```

```
    static Color generateRandomColor();
```

```
}
```

```
class Shape2D {
```

```
protected:
```

```
    Color color;
```

```
    Visibility visibility;
```

```
public:
```

```
    Shape2D(const Color &color);
```

```
    ~Shape2D();
```

```
    void setVisibility(Visibility visibility);
```

```
    virtual bool inBounds(float canvasWidth, float canvasHeight) = 0;
```

```
    virtual void draw() = 0;
```

```
}
```

```
class Text: public Shape2D {
```

```
private:    float x, y;    unsigned int fontSize;    string text;
```

```
public:
```

```
    Text(Color color, float x, float y, uint fontSize, string text);
```

```
    bool inBounds(float canvasWidth, float canvasHeight) override;
```

```
    void draw() override;
```

```
}
```



```
class Rectangle : public Shape2D {
```

```
private:
```

```
    float x, y, width, height;
```

```
public:
```

```
    Rectangle (Color color, float x, float y, float width, float height);
```

```
    bool inBounds (float canvasWidth, float canvasHeight) override;
```

```
    void draw() override;
```

```
}
```

```
class Canvas {
```

```
private:
```

```
    float width, height;
```

```
    vector<Shape2D*> shapes;
```

```
public:
```

```
    Canvas (float width, float height);
```

```
    ~Canvas ();
```

```
    bool addShape (Shape2D* shape);
```

```
    void resize (float width, float height);
```

```
    void draw();
```

```
    void undo();
```

```
}
```